

Pulsed N M R – The Buzz

Larmor precession (Ref. 1, pp 2-3)

A **nuclear magnetic moment** μ is associated with **non-zero nuclear spin** I (= nuclear angular momentum) according to:

$$\mu = \gamma \hbar I / 2\pi \quad \dots (1)$$

where γ is the **gyromagnetic ratio** of the nucleus.

When placed in a magnetic field $\mathbf{B}_0$, in order to conserve angular momentum, a nuclear magnetic moment will **precess** about the magnetic field. This **Larmor** precession is described by the equation

$$\omega_0 = -\gamma \mathbf{B}_0 \quad \dots (2)$$

where ω_0 is the **angular** Larmor frequency (related to the **cyclic** frequency f_0 by $\omega_0 = 2\pi f_0$, as usual).

Question 1: The Larmor frequency f_0 , of hydrogen protons (^1H), in our PS2 pulsed NMR spectrometer, is (depending on the room temperature) close to 21.7 MHz. What is the magnitude of the magnetic field B_0 (in T) set up by the permanent magnet at the sample position?

Answers: A) 0.4 B) 0.5 C) 0.6 D) 0.7

Question 2: What do you expect the Larmor frequency of ^{19}F nuclei to be in this spectrometer (in MHz)?

Answers: A) 19.5 B) 20.4 C) 21.1 D) 23.1

Oscillating magnetic fields, the rotating frame of reference and RF pulses (Ref. 1, pp 4-5)

A magnetic field $\mathbf{B}_1(t) = 2 B_1 \mathbf{j} \sin \omega t$, oscillating in the y-direction ($\mathbf{j}$ is the y unit vector) at a frequency ω , can be written as the sum of two rotating components, each with **half the amplitude** of the oscillating field and rotating clockwise and anti-clockwise, respectively i.e.:

$$\mathbf{B}_1(t) = B_1(\mathbf{j} \sin \omega t - \mathbf{i} \cos \omega t) + B_1(\mathbf{j} \sin \omega t + \mathbf{i} \cos \omega t) \quad \dots (3)$$

where $\mathbf{i}$ is the x unit vector.

In NMR, the field component rotating in the same sense as the nuclear precession can cause transitions between the nuclear energy states, while the component rotating in the opposite sense has very little effect and is ignored.

We now change the view point, from our laboratory frame of reference (x, y, z) to a frame of reference (x', y', z) which is rotating at the Larmor frequency ω_0 , about the z-axis (= the direction of $\mathbf{B}_0$). This rotation effectively cancels out the steady magnetic field $\mathbf{B}_0$ applied in the lab frame, while the $\mathbf{B}_1$ field component, rotating at ω_0 , reduces to a **static field** in the rotating frame i.e. the effective magnetic field becomes:

$$\mathbf{B}_{\text{eff}} = B_1 \mathbf{j}' \quad \dots (4)$$

where $\mathbf{j}'$ is the rotating y' unit vector (Ref. 1, pp 4-5).

In the rotating frame of reference, the motion of a nuclear magnetic moment μ , will then be a **precession about the y' -axis**, with an angular frequency $\omega_1 = -\gamma B_1$ (c.f. Eqn. (2)).

If the rotating field B_1 is **pulsed**, i.e. is switched on for a short time τ and is then switched off again, the nuclear moment will, during the 'on' time, precess by an angle $\theta = \tau\omega_1 = -\tau\gamma B_1$. For a given B_1 (Note: we cannot vary B_1 in the PS2) we can adjust τ so that $\theta = \pi/2$ – this is called a 90° pulse and means that the magnetic moment will precess 90° in the $x'z$ -plane. Doubling the time the pulse is switched on for, then gives us a magnetic moment precession of 180° (i.e. an inversion of the magnetic moment) and is called a 180° pulse.

Question 3: If in our PS2 spectrometer a 90° pulse is about $2.50\ \mu\text{s}$ long, what is the RF field amplitude (in tesla) generated by the RF coil at the sample position?

Answers: A) 2.35×10^{-3} B) 2.35 C) 0.5×10^{-3} D) 0.5

Question 4: If a 90° pulse is about $2.50\ \mu\text{s}$ long in the PS2, approximately how much will a $1.0\ \mu\text{s}$ RF pulse rotate the hydrogen nuclear moments?

Answers: A) 1° B) 2.5° C) 18° D) 36°

Relaxation

(i) Spin – lattice relaxation, T_1

At a given nucleus, a spectrum of **time varying** magnetic fields is produced, by the **thermal motion** of magnetic moments of neighbouring nuclei and electrons. Those field components, which **happen** to oscillate at the Larmor frequency, will stimulate transitions of nuclei between their energy states.

Transition of nuclei to states of lower energy is called spin-lattice (or **longitudinal**) relaxation and is described by the **spin-lattice relaxation time, T_1** .

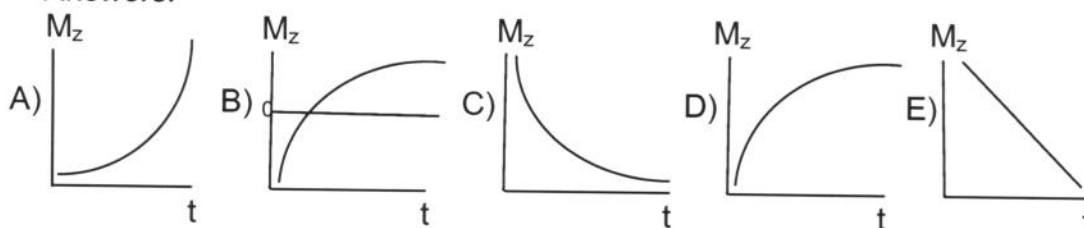
After applying a 90° RF pulse, the nuclear magnetisation in the z -direction $M_z(t)$, is expected to recover with time t , according to:

$$M_z(t) = M_0(1 - \exp(-t/T_1)) \quad \dots (5)$$

where M_0 is the equilibrium value of the z -magnetisation.

Question 5: The magnetisation (M_z) in Eqn. (5) above looks like?

Answers:



Question 6: If the T_1 of hydrogen nuclei in a sample is 60 ms, what fraction of the nuclei will have relaxed back to the B_0 direction 60 ms after applying a 90° pulse?

Answers: A) 0.693 B) 0.632 C) 0.368 D) 0.306

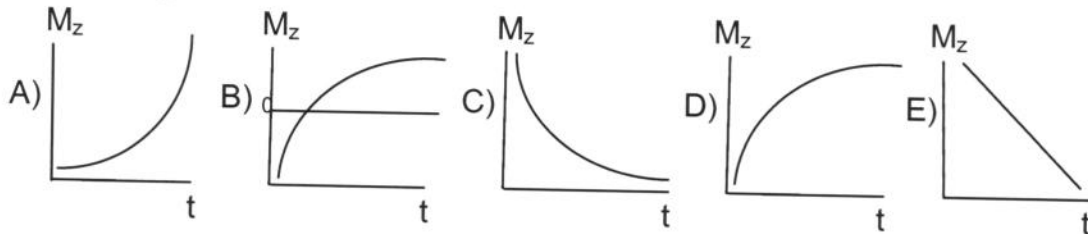
After applying a 180° RF pulse, the nuclear magnetisation in the z-direction $M_z(t)$, is expected to recover with time t , according to:

$$M_z(t) = M_0(1 - 2 \exp(-t/T_1)) \quad \dots (6)$$

where M_0 is the equilibrium value of the z- magnetisation.

Question 7: The magnetisation (M_z) in Eqn. (6) above looks like?

Answers:



Question 8: In Eqn. (6) what is the ratio of M_z / M_0 at $t = 0$?

Answers: A) 2 B) 1 C) 0 D) -1 E) -2

Question 9: In Eqn. (6), what is the ratio t / T_1 when $M_z = 0$?

Answers: A) 1.414 B) 0.1.000 C) 0.707 D) 0.693 E) 0.368

Measurements of signal amplitude (A – volts) versus time (t – seconds) are least-squares fitted with an equation of the form $A(t) = A_0(1 - 2 \exp(-t/\tau))$, where A_0 and τ are the fit parameters.

Question 10: The units of A_0 are?

Answers: A) V B) V^{-1} C) s D) s^{-1}

Question 11: The units of τ are?

Answers: A) V B) V^{-1} C) s D) s^{-1}

(ii) Spin – spin relaxation, T_2

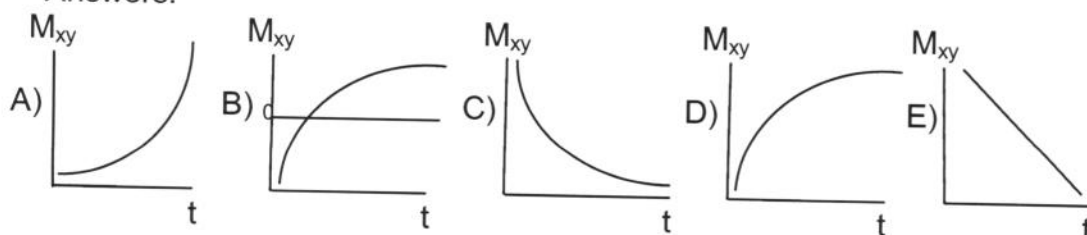
A 90° pulse rotates nuclear magnetic moments into the xy-plane, resulting in a nuclear magnetisation M_{xy} . Neighbouring nuclei and electrons surrounding a given nucleus can set up a net magnetic field at that nucleus. This field will be slightly different from the fields at other nuclei in the sample. These slightly different fields, seen by each nucleus, mean that a time t after a 90° pulse has been applied, the nuclear magnetic moments will get out of phase with each other according to:

$$M_{xy}(t) = M_0 \exp(-t/T_2) \quad \dots (7)$$

where the **spin-spin** (or **transverse**) **relaxation time T_2** , is the time constant which describes the decay of the transverse components of the magnetisation.

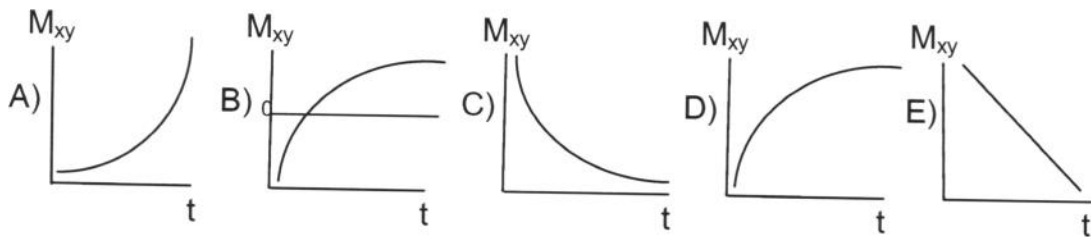
Question 12: The magnetisation (M_{xy}) in Eqn. (7) above looks like?

Answers:



Question 13: If the magnetisation (M_{xy}) in Eqn. (7) is plotted using a logarithmic scale, the curve would look like?

Answers:



Question 14: If the spin-spin relaxation time (T_2) of the hydrogen nuclei in a sample is 50 ms, what fraction of the nuclei would still be left in the xy-plane 50 ms after applying a 90° pulse?

Answers: A) 0.693 B) 0.632 C) 0.368 D) 0.306

Free Induction Decay (FID)

After a 90° pulse the magnetisation M_{xy} , which has been rotated into the xy plane, is expected to de-phase according to Eqn. (7), purely due to spin-spin relaxation. In practice the applied field B_0 would be expected to be slightly inhomogeneous (i.e. different in different parts of the sample). This would result in the xy nuclear magnetisation de-phasing somewhat faster, according to:

$$M_{xy}(t) = M_0 \exp(-t / T_2^*) \quad \dots (8)$$

where $T_2^* (< T_2)$ includes a contribution due to the field inhomogeneity.

In the PS2, the NMR signal is detected by the same coil which supplies the pulsed RF field in the y-direction. The resulting signal, as the nuclear magnetisation rotates in the xy-plane and the nuclear moments de-phase, is called the **Free Induction Decay (FID)**. In the FID, the oscillations at the Larmor frequency decay exponentially as shown in Fig. 1.

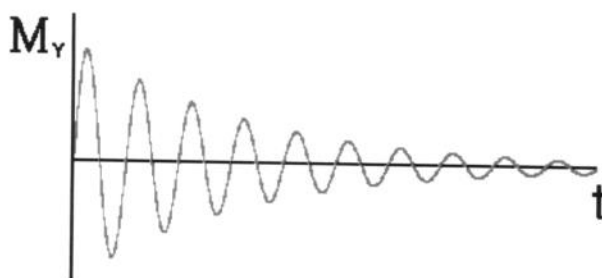


Figure 1: Free induction decay after application of a 90° pulse

Question 15: Changing T_2^* in Eqn. (8) would change the FID's (Fig. 1):

A) frequency B) amplitude C) decay time D) time period

The FID signal is rectified and its envelope is output at the 'Env. Out' socket of the '21 MHz Receiver' module. When displayed on the DSO screen the trace looks like an exponentially decaying voltage (Fig. 2).

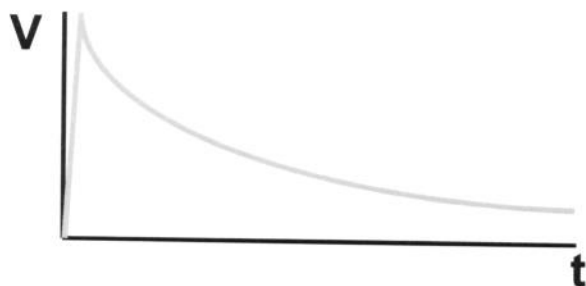


Figure 2: DSO screen trace of the FID appearing at 'Env. Out'

The signal appearing at the 'I Out' socket of the '21 MHz Receiver' module is the Larmor frequency signal of Fig.1 multiplied by a square wave generated at the spectrometer frequency, set by you on the '21 MHz Synthesizer'. The result is then a decaying signal, oscillating at the difference (**beat**) frequency between the Larmor and the spectrometer frequencies. Adjusting the 'Synthesizer' frequency to give **zero beats** then sets the '21 MHz Synthesizer' frequency **equal** to the Larmor frequency!

Question 16: If the Larmor frequency of hydrogen nuclei in a particular sample in the spectrometer is 21.69600 MHz and the PS2 'Synthesizer' is set to 21.70000 MHz, what would be the oscillation period (in ms) of the beats appearing at 'I Out'?

Answers: A) 0.4 B) 2.5 C) 0.04 D) 0.25 E) 4.0

Chemical shift

Nuclei in an atom/molecule are partially screened from the full effect of the applied magnetic field B_0 , by the surrounding electrons. The field 'seen' by a particular nucleus is then $B = B_0 (1 - \sigma)$ where σ is called the **chemical shielding** constant. The electron density, and hence σ , vary slightly ($\sim$ parts per million for ^1H nuclei) for the same nuclei located in different parts of a molecule, i.e. in different **chemical groups** within a molecule. This is called the **chemical shift** and is widely used in the determination of molecular structure in chemistry.

Question 17: Two peaks in the ^1H NMR spectrum of a molecule occur at 20.0000 MHz and 20.0001 MHz. What is the chemical shift **difference**, expressed in parts per million (ppm), between these peaks?

Answers: A) 100 B) 50 C) 20 D) 10 E) 5

Fourier Transforms

Time varying functions can be described either by showing their evolution in time or by specifying their constituent frequency (= **reciprocal** of time) components. This is an example of a Fourier transform (FT) which, in general, transforms the description of a function from one space to a description via variables in a reciprocal space.

Because of this reciprocal nature 'narrow' functions transform to 'wide' functions and vice-versa.

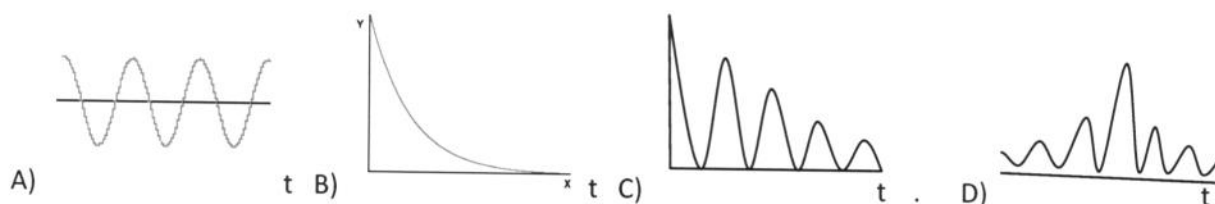
Question 18: If the Fourier transform of a Gaussian voltage pulse of width 20 ms, is a Gaussian of width nearly 500 Hz, what will be the width (in Hz) of the Fourier transform of a Gaussian pulse 10 ms wide?

Answers: A) 2000 B) 1000 C) 500 D) 200 E) 100

In the present experiment the DSO can calculate and display the (digital) Fourier transform of a time dependent FID signal, in order to find its constituent frequency components. A 'narrow' peak on the FT display then corresponds to a narrow range of oscillation **frequencies**.

Question 19: If the two peaks of Question 17 are of equal intensity, what would you expect the FID (= the Fourier Transform) of the two peaks to look like?

Answers:



Magnetic field gradients

We now apply a **linear magnetic field gradient** G_y in the y-direction, to our sample. The applied magnetic field B_z will no longer be the same for all the nuclei in a sample of length L in the y-direction, but will vary with position according to:

$$B_z(y) = B_0 + c y G_y \quad \dots (9)$$

for y between 0 and L (c is a constant); $B_z = B_0$ otherwise.

This in turn means that the **Larmor frequency of the nuclei will vary linearly in the y- direction** in the region $0 < y < L$. Hence, the Fourier transform (= frequency) trace, rather than just showing a narrow peak around a single frequency (Fig. 3 a), will show a 'plateau' (Fig. 3 b) corresponding to a **distribution** of peaks over the **range of frequencies** derived from Eqn. (9).

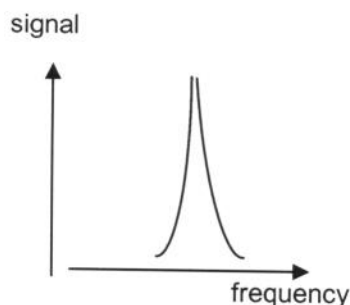


Fig. 3 a) no field gradient

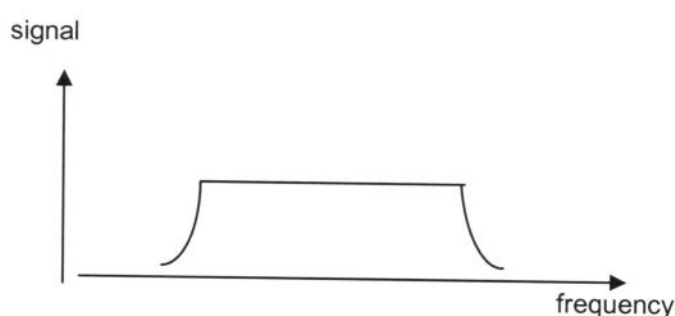


Fig. 3 b) with field gradient

The plateau width (in frequency units) then corresponds to the length L of the sample in the y-direction, giving us a mapping between spatial position in the y-direction and the Larmor frequency of the nuclei.

Question 20: An oil sample 5 mm long gives a 'plateau' of width 1000 Hz when a field gradient is applied in the vertical (= y) direction. If a small air bubble (which gives zero NMR signal) is present in the oil and gives a dip in the 'plateau' signal about 200 Hz away from the 'plateau' centre, how far (in mm) is the bubble vertically from the sample's centre?

Answers: A) 1 B) 2 C) 3 D) 4 E) 5

Magnetic Resonance Imaging - MRI

Different bodily tissues (e.g. bone, muscle, fat, tumours etc.) have different proton densities and different relaxation times. MRI allows us to distinguish between the positions of various types of tissue, by applying suitable field gradients to bodily parts immersed in a steady field, B_0 .

Glossary

90° pulse - a pulse of oscillating magnetic field, at the Larmor frequency, just long enough to rotate the nuclear magnetisation through 90°

180° pulse - a pulse of oscillating magnetic field, at the Larmor frequency, just long enough to rotate the nuclear magnetisation through 180°

chemical shielding – the partial shielding of a nucleus, from the static applied magnetic field, by surrounding atomic electrons,

chemical shift – the difference in chemical shielding, in various parts of a molecule, resulting in somewhat different Larmor frequencies for nuclei in different molecular positions

Fourier transform – transformation of the description of a function from one space (e.g. time) to a description via variables in a reciprocal space (frequency).

free induction decay – the decaying, oscillating (at the Larmor frequency) signal which follows application of a 90° rf pulse to nuclei

gyromagnetic ratio – the ratio of the nuclear magnetic moment of a nucleus to the spin angular momentum of the nucleus

inhomogeneity – unevenness with respect to change in spatial position

inversion recovery – the recovery of the nuclear z-magnetisation (M_z), with time, after inversion by a 180° pulse

Larmor precession – the precession of a nuclear magnetic moment about a steady magnetic field

magnetic field gradient – the variation of a magnetic field as a function of spatial position

magnetic resonance imaging (MRI) – an imaging technique employing NMR with applied magnetic field gradients, to differentiate signals from various sections of a sample

nuclear spin – the angular momentum of a nucleus expressed in units of $\hbar/2\pi$

nuclear magnetic moment - the magnetic moment of a nucleus, usually given in nuclear magnetons

precession - rotation of the axis of angular momentum of a nucleus about a magnetic field

pulsed oscillating magnetic field – an oscillating magnetic field switched on for a short time

rf magnetic field – a magnetic field oscillating in the MHz frequency region

rotating frame of reference – in NMR, the frame of reference rotating at the Larmor frequency about the applied magnetic field direction

spin echo - signal from the re-phasing of nuclear spins in the x-y plane, after the application of a 90° - 180° rf pulse sequence

spin-lattice relaxation – the return of the z nuclear magnetisation (M_z) to its equilibrium value along the z axis

spin-spin relaxation – the return, with time, of the xy nuclear magnetisation (M_{xy}) to zero

static magnetic field - a magnetic field which does not change with time

REFERENCES

- 1) "Magnetic resonance spectroscopy and imaging"
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- 3) <http://www.cis.rit.edu/htbooks/nmr/inside.htm>
- 4) <http://www.cis.rit.edu/htbooks/mri/inside.htm>